Zendo chain white paper

Version 0.9.0 · Software description, not an offer of securities

This note describes the zendo-1 software in this repository. Balances are ZendoCoin on the ledger the node writes. They are not dollars. Whether an operator needs a money-transmitter or other license depends on what they do and where they do it. This document is not legal, financial, or investment advice.

What runs

One Next.js process serves the site, the wallet, the explorer, and a JSON ledger at `data/state.json`. A separate Rust binary, `zendo-node`, keeps its own blocks, gossip, and a small stack machine. The website does not send wallet transfers through that binary. Two data directories are two chains unless you enroll a second validator into the same genesis before the first block.

ZendoCoin

Ticker $ZCoin. Base supply is 1,000,000,000 with 18 decimals. Founder locks hold 300,000,000: Louis Conner, Co-founder and President, 200,000,000; Prabhat Agrawal 50,000,000; Pratyaksh Lutare 50,000,000. The public allocation is 700,000,000 and is liquid from genesis. It includes the system wallet seed of 100,000, the grant treasury, the launchpad studio, and the remaining public pool. New accounts receive 0 ZCoin. The faucet does not pay at signup. Staking rewards are the only inflation. They are minted when a delegator claims, at the configured annual rate, after the validator commission. Jailing a validator slashes 10 percent of its self-stake and of each delegation into the treasury, and pauses rewards until it is active again. The slash is not refunded. After genesis, the node does not invent trades.

Supply stays conserved: spendable balances, validator self-stake, delegations, and unbonding equal the base supply plus rewards that have already been issued. The node checks that sum before it writes state.

Founder vesting

Each founder has a vesting account and a separate wallet, both addresses beginning with znd1. The vesting clock is not set until this app is first served on a public HTTPS host. That instant is stored once and is not a date chosen in advance. Nothing unlocks before that launch, and nothing unlocks during the 120-day cliff after it, including on the cliff day itself. After the cliff, 25 percent of that founder’s own allocation unlocks at each anniversary for four years. The chain moves those coins into that founder wallet by itself. No operator has to start the clock or press a claim. It does not credit the system wallet. Coins that are still locked stay in the vesting account.

Launchpad

A new coin is created for a fee of 0.2 ZCoin. The creator does not receive the supply. Price follows a constant-product curve against virtual reserves. Each trade pays 1 percent: 30 basis points to the creator and 70 basis points to the protocol. Gas is charged on top. When real ZCoin reserves reach 85, the curve graduates. Raised ZCoin and the remaining tokens seed an automated pool, and later trades use that pool. Promoted slots cost 50 ZCoin for 24 hours. Optional extra protocol fee of 5 basis points is off unless the config turns it on. That switch does not reorder other people's transactions.

Governance and grants

A proposal costs 1 ZCoin and needs 100 ZCoin of voting power to open. It names one allowed parameter. A vote weighs liquid ZCoin plus delegated stake, snapshotted when you vote, and costs 0.002 ZCoin. Quorum is 1,000 ZCoin. A passing vote writes that parameter into the ledger, and later blocks use it. A vote cannot change the base supply, the 18 decimals, or the vesting schedule. Grants follow the same power check, lock for 60 seconds on a live node, and pay from the treasury only when the yes side leads and quorum is met. Marketplace listings are transactions on the same ledger.

Accounts

Signup uses email and a password stored as a bcrypt hash. The wallet address comes from a 12-word BIP39 phrase. The server stores a hash of the phrase, not the words. A 4-digit PIN, also hashed, is required after the password. Five wrong PINs pause entry for five minutes. Password reset links are shown on screen because this build has no mail server. Recovery with the email and the 12 words replaces the password and the PIN. Sessions are signed cookies. Production marks them secure and refuses public HTTP. Loopback health checks stay open. Public JSON-RPC is rate limited at 60 calls a minute, or 600 for a premium account.

The Rust node

Configure `zendo-node` from the command line. The website reads `zendo-blockchain.toml` for the genesis fees, the validator table, and the chain id `zendo-1`. Passed votes overlay those fees in the ledger. The Rust process speaks TCP gossip, prefers peers with a lower measured round trip, chooses the longer chain when the common ancestor is still above the finalized height, and meters gas on a stack machine. Programs in the crate cover a token, cliff vesting, a curve, a ballot, an NFT, and an identity registry. The deployer can replace code or roll it back. Equivocation slashes 10 percent once per height. Staking bonds sit at a fixed address. JSON-RPC has no authentication, so keep it on localhost.

Fees

Gas on transfers and votes, the bonding-curve fee, creation and promotion fees, and staking inflation are the revenue surfaces in this build. Premium RPC raises the free limit of 60 calls a minute to 600. There is no card processor, no fiat payout, and no separate custodial balance.

End of version 0.9.0.